11 Graph Theory

This is from Chapter 11 of the text.

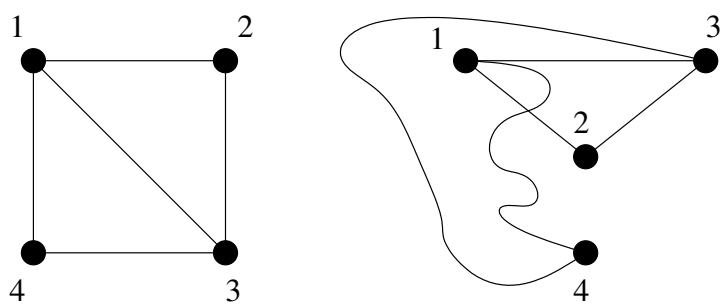
11.1 Introduction to Graph Theory

Definition 11.1. A graph $G = (V, E)$ consists of a non-empty set V of *vertices* and a set E of 2-element subsets of V, called *edges*.

For example

$$G = (\{1, 2, 3, 4\}, \{\{1, 2\}, \{2, 3\}, \{3, 4\}, \{4, 1\}, \{1, 3\}\})$$

is a graph. It has four vertices and five edges. As this mess of brackets gets troublesome to write, we drop them when it doesn't confuse things an write an edge $\{x, y\}$ as xy. The graph G can be represented pictorially as



We often write $V(G)$ for V and $E(G)$ for E if we don't define the these sets for a graph G explicitly.

There are a lot of definitions related to the fact that $e = uv$ is an edge of G. We say

- the vertices u and v are *adjacent*, written $u \sim v$,
- u and v are *neighbours*,
- u and v are the *endpoints* of e,
- u (and v) is incident with e,

Note

One can generalise the idea of a graph to

- *digraphs* by making edges ordered sets: uv and vu are different,
- *multigraphs* by making the edgeset E a multiset,
- *hypergraphs* by allowing edges to contain more than two vertices,
- *non-simple graphs* by allowing edges of the form uu.

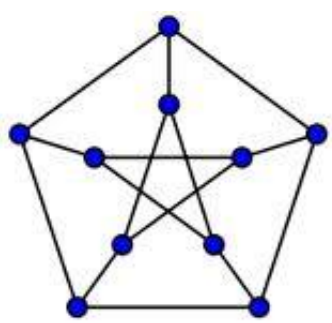
We mostly don't consider these generalisations.

We used graphs when we considered the problem of Ramsey:

> How many people must there be at a party so that we can be sure that there are three mutual acquaintences, or three mutual strangers?

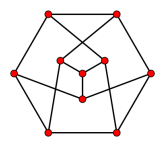
We represented the people at the party as vertices of a graph, and the relationship of 'begin acquainted' by edges. This is just one example of how graphs can be used to model problems. Graph theory studies the properties that arise in structure due to the simple idea of a relationship between elements of a set. Many such properties are purely combinatorial, others have more structural ramifications, which arise surpisingly in most any field of mathematics.

Here is a graph, called the *Petersen Graph*, and denoted P, that some people like to put on t-shirts, and refer to fondly as the *universal counter-example*:



Here are some typical questions that we might ask of this, or any graph. The definitions you need to understand these questions come later, but they are probably what you think they are.

- How many colours do we need, when colouring the vertices of P to be sure that adjacent vertices get different colours.

- Can you draw P without intersecting edges?
- How many edges (vertices) must you remove to *disconnect* P?
- For any two pairs of vertices u, u' and v, v' can you find a *path* between u and u', and another between v and v', such that the two are disjoint?
- What is the smallest *cycle* in P?
- How many 5-cycles are there in P?
- Can we partition the edges of P into disjoint 5-cycles?
- Can we find a *walk* in P that visits every vertex (edge) exactly once?
- Is P the same graph as this: 

11.1.1 Isomorphisms

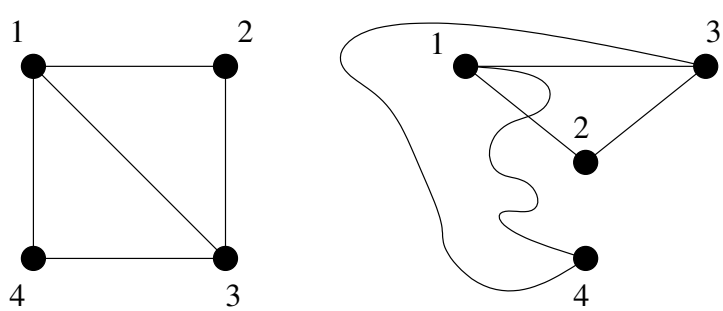
Often we don't care about the names of the vertices, and want to consider graphs to be the same if they look the same when ignoring these vertex names. For this we make the following defintion.

Definition 11.2. An *isomorphism* is a bijective map $f : V(G) \to V(H)$ such that

$$uv \in E(G) \iff f(u)f(v) \in E(H).$$

If there is an isomorphism f from G to H, we say G and H are *isomorphic*, and write $G \cong H$.

Example 11.3. The following graphs are all isomorphic.



It is a hard problem to decide if two graphs are isomorphic. We talk about this a little more later.

Theorem 11.4. *Isomorphism is an equivalence relation. That is:*

i. $G \equiv G$ *(symmetry)*

ii. $G \equiv H \Rightarrow H \equiv G$ *(reflexivity)*

iii. $G \equiv H$ *and* $H \equiv J \Rightarrow G \equiv J$ *(transitivity)*

Problem 11.1. Prove this!

Problem 11.2. How many graphs are there on 4 vertices? How many non-isomorphic (unlabelled) graphs?

As we usually consider graphs to be the same if they are isomorphic, we would like to decide if two graphs are isomorphic. Deciding if two graphs are isomorphic is usually pretty hard (this statement can be made rigourous). But if we are lucky, there are easy ways to prove that they are not.

For example, if two graphs have a different number of vertices, or a different number of edges, they are not isomorphic. Such properties, which are 'preserved by isomorphism', are called graph properties. Graph Theory is the study of these properties.

11.1.2 Some Common Graphs

We generally now consider graphs to be the same if they are isomorphic, and take blatent liberties with the names of their vertices. As such, we define some common graphs, but will apply these names to any graph that is isomorphic.

- The complete graph on n vertices, K_n, has $V(K_n) = \{v_1, v_2, \dots v_n\}$ and $E(K_n) = \{v_i v_j \mid 1 \leq i < j \leq n\}$.
- The n-cycle, C_n, has $V(C_n) = \{v_1, v_2, \dots v_n\}$ and $E(C_n) = \{v_i v_{i+1} \mid i = 1, \dots, n-1\} \cup \{v_n v_1\}$.
- The n-path, P_n, has $V(P_n) = \{v_0, v_1, \dots v_n\}$ and $E(P_n) = \{v_i v_{i+1} \mid i = 0, \dots, n-1\}$.
- The complete bipartite graph, $K_{m,n}$, has $V(K_{m,n}) = \{a_1, \dots a_m\} \cup \{b_1, \dots, b_n\}$ and $E(K_{m,n}) = \{a_i b_j \mid i \in [m], j \in [n]\}$.
- The n-cube, Q_n, $V(Q_n)$ is the set of binary strings of length n, vertices u and v are adjacent if they differ in exactly one coordinate.

11.1.3 Homomorphisms

Definition 11.5. A *homomorphism* $f : G \to H$ of a graph G to another graph H, is a vertex map $f : V(G) \to V(H)$ such that

$$uv \in E(G) \Rightarrow f(u)f(v) \in E(H).$$

If there is a homomorphism $f : G \to H$, we write simply $G \to H$.

Any isomorphism is a homomorphism, and as usual, a homomorphism $f : G \to H$ is an isomorphism if and only if there is an inverse homomorphism $f^{-1} : H \to G$ such that $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity maps. But there is another property you may be used to for other finite structures, that doesn't hold for us: **a bijective homomorphism is not necessarily an isomorphism.** Indeed, there are many bijections that are not isomorphisms.

Example 11.6. For any graph G on n vertices there is a bijective homomorphism $f : G \to K_n$.

A homomorphism $f : G \to H$ is a map on the vertices, but it induces a map $f_E : E(G) \to E(H)$ on the edges:

$$f_E(uv) = f(u)f(v).$$

The homomorphism f is called *edge-injective* if f_E is injective, and *edge-surjective* if f_E is surjective.

Problem 11.3. Show that if f is injective, then it is edge-injective, but that the converse is not true.

Problem 11.4. Show that f is an isomorphism if and only if it is bijective and edge-surjective.

Problem 11.5. Show that $C_{2n+1} \to C_{2m+1}$ if and only if $m \leq n$. Show that $C_n \to K_2$ if and only if n is even. Show that $G \to K_2 \Rightarrow C_{2n+1} \not\to G$ for every n.

11.2 Subgraphs, Complements, and Degree Sequences

Given a graph G, another graph G' is a *subgraph* of G, written $G' \leq G$ if $V(G') \subset V(G)$ and $E(G') \subset E(G)$.

A subgraph G' of G is *spanning* if $V(G') = V(G)$, or *induced* if $E(G') = \{uv \in E(G) | u, v \in V(G')\}$. A subgraph G' of G is a *proper* subgraph of G if $E(G')$ is a proper subset of $E(G)$.

Problem 11.6. How many non-labelled subgraphs are there of K_4? How many are proper? ...induced? ...spanning? ...spanning, induced and proper?

Definition 11.7. The *complement* $\overline{G}$ of a graph G is the graph defined by $V(\overline{G}) = V(G)$ and $E(\overline{G}) = \overline{E(G)}$, (where the universe used for complementation is the set $\{uv \mid u \neq v \in V(G)\}$ of all possible edges on $V(G)$).

Problem 11.7. Show that a subgraph G' of G is induced if and only if $\overline{G'}$ is a subgraph of $\overline{G}$.

Solution

Let $G' \subset G$ be induced. Clearly $V(\overline{G'}) \subset V(\overline{G})$, so we have to show that $E(\overline{G'}) \subset E(\overline{G})$. Let $e \in E(\overline{G'})$ then $e \notin E(G')$ so $e \notin E(G)$. So $e \in E(\overline{G})$.
On the other hand, let $\overline{G'} \leq \overline{G}$. We want to show that G' is induced. For $u, v \in V(G')$, with $uv \in E(G)$ we want to show that $uv \in E(G')$. Well $uv \in E(G)$ implies that $uv \notin E(\overline{G})$ so not in $E(\overline{G'})$, so because $u, v \in V(G')$, $uv \in E(G)$.

Problem 11.8. Show that a bijective homomorphism $f : G \to H$ is an isomorphism if and only if it is also a homomorphism of $\overline{G} \to \overline{H}$.

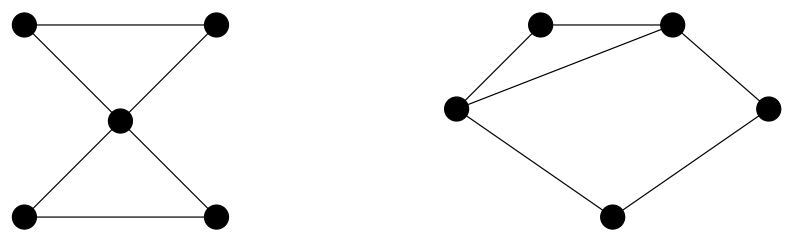
Problem 11.9. A graph G is self-complementary if $G \equiv \overline{G}$. How many edges are there in a self-complementary graph on n vertices?

11.2.1 Degrees and Degree Sequences

Definition 11.8. For a vertex v in a graph G, the *degree* of v, written $\deg(v)$, is the number of edges in G incident to v. The *degree sequence* of a graph G is the multiset of the degrees of the vertices of G. The minimum degree, $\delta(G)$, and maximum degree, $\Delta(G)$, of a graph G are the minimum and maximum degrees respectively in its degree sequence.

Problem 11.10. Show that every graph contains a path of length at least $\delta(G)$.

Clearly, degree sequence, minimum degree, and maximum degrees are graph properties. So the following two graphs, which have the same number of vertices, and edges are not isomorphic as they have different degree sequences: $(4, 2, 2, 2, 2)$ and $(3, 3, 2, 2, 2)$ respectively, (and in fact different maximum degrees.)



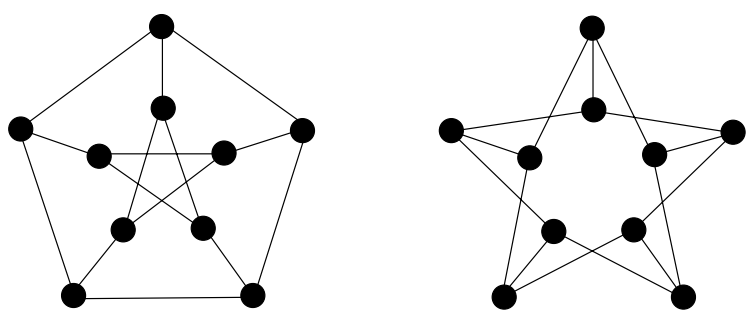
Here is a simple observation about degree sequence:

Theorem 11.9. *If $(d_1, d_2, \ldots, d_n)$ is the degree sequence of a graph, then the degree sum: $\sum_{i=1}^{n} d_i$ is equal to $2|E(G)|$.*

Proof. Each edge of the graph contributes 2 to the degree sum, one for each endpoint. □

Corollary 11.10. *In any graph G, there is an even number of vertices of odd degree.*

Problem 11.11. What are the degree sequences of the following graphs? Are the following two graphs, with the same degree sequences, isomorphic?



Problem 11.12. What is the degree sequence of the cube Q_n?

Definition 11.11. A graph G is called *k-regular* if every vertex has degree k.

Problem 11.13. How many edges does a k-regular graph on n vertices have?

A sequence $(d_1, d_2, \ldots, d_n)$ non-increasing integers is *graphic* if it is the degree sequence of a graph. We saw above that one necessary condition for a sequence

to be graphic is that it sums to an even number. But this is not sufficient: $(4, 2, 0)$ is not graphic.

Problem 11.14. Show that the following condition is also necessary for a sequence $(d_1, d_2, \ldots, d_n)$ to be graphic. For all $1 \leq k \leq n$,

$$\sum_{i=1}^{k} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

In fact, this condition is also sufficient. But this is fair bit harder to show:

Theorem 11.12. *(Erdős-Gallai '60) A sequence $(d_1, d_2, \ldots, d_n)$ is graphic if and only if $\sum_{i=1}^{k} d_i$ is even and for all $1 \leq k \leq n$,*

$$\sum_{i=1}^{n} d_i \leq k(k-1) + \sum_{i=k+1}^{n} \min(k, d_i).$$

The Erdős-Gallai condition is what we use to decide if a sequence is graphic, but there is also a recursive method to decide it. It depends on the following idea:

Lemma 11.13. *(Havel-Hakimi) If a sequence $D = (d_1, d_2, \ldots, d_n)$ is graphic, then there is a graph with degree seqence D such that the maximum degree vertex is adjacent to the d_1 vertices of next highest degree.*

Thus to decide if a sequence D, such as $(3, 3, 2, 2, 2)$, is graphic, the *Havel-Hakimi algorithm* is as follows. While D is not all zeros,

- remove d_1, and
- remove 1 from the next d_1 degrees $d_2, \ldots, d_{d_1-1}$,

if possible.

The algorithm clearly terminates. If it terminates with an empty sequence, (or a sequence of zeros), then D is graphic. Otherwise, it is not graphic.

Starting with $(3, 3, 2, 2, 2)$, we get $(\cancel{3}, 3-1, 2-1, 2-1, 2) = (2, 2, 1, 1)$, and then $(1, 0, 1) = (1, 1)$, and then $(0, 0)$. So $(3, 3, 2, 2, 2)$ is graphic.

After one definition, let's prove the Lemma.

Definition 11.14. The *neighbourhood* $N(v)$ of a vertex v in a graph G is the set

$$\{u \in V(G) \mid uv \in E(G)\}$$

of all neighbours of v.

Proof. Let $D = (d_1, d_2, \ldots, d_n)$ be a graphic sequence. Given a realisation G of D, let v be a vertex of degree $\Delta = d_1$, and let $T = (v_2, \ldots, v_{\Delta-1})$ be a set of Δ vertices where v_i has degree d_i for all i. If $N(v) = T$ then we are done, so towards contradiction, assume that the realisation G of D that maximises $\Delta' = |N(v) \cap T|$ has $\Delta' < \Delta$, and assume that G is such a realisation.

We find a new realisation G' with $|N(v) \cap T| > \Delta'$, so the lemma follows by contradiction.

As $N(v) \neq T$, there is $t \in T$ with $v \not\sim t$ and $s \notin T$ with $v \sim s$. Furthermore, since $t \in T$ and $s \notin T \cup \{v\}$, $\deg(t) \geq \deg(s)$. So there is some vertex w with $w \sim t$ and $w \not\sim s$.

Removing the edges vs and tw from G and adding vt and sw we get a graph G' with the same degrees on every vertex, (so another realisation of D), but with $|N_{G'}(v) \cap T| = |(N_G(v) \cap T) \cup \{t\}| = \Delta' + 1$, as needed for the contradiction. $\square$

Problem 11.15. Decide, using Havel-Hakimi, whether or not the sequence $(5, 5, 5, 3, 2, 2, 1, 1)$ is graphic.

11.3 Walks, trails, paths, circuits and cycles

Definition 11.15. Let x and y be vertices in a graph G. An xy-walk is a finite alternating sequence of vertices and edges:

$$x = x_0, e_1, x_1, e_2, e_2, \ldots, x_{n-1}, e_n, x_n = y$$

where $e_i = x_{i-1}x_i$ for $i = 1, \ldots n$.

Usually we just write the vertices

$$x_0 x_1 \ldots x_n,$$

or, if x and y are known, the edges

$$e_1 e_2 \ldots e_n.$$

A walk in a graph G can be viewed as a homomorphism of a path P_n to G. The *length* of the walk is n, the number of edges in it. If a walk is edge-injective, it is called a *trail*; if it is injective, it is called a *path*.

An xy-walk is *closed*, or a *circuit* if $x = y$. This can be viewed as a homorphism of a cycle C_n to G. If it is injective, it is called a *cycle*. The *girth* of the cycle is n.

Theorem 11.16. *Any xy-trail in a graph has an xy-path as a subgraph.*

Intuitively, we just start with an xy-trail an 'chop' out the repeated bits until it is a path. A lot of times it is very hard to write the intuitive proof rigorously. Try it before you proceed. A nice way to prove this theorem is the following proof by contradiction, using a minimum counterexample.

Proof. By assumption, there exists an xy trail in G. Let $x = x_0, x_1, \dots x_d = y$ be the shortest xy trail.

Towards contradiction, assume that the trail isn't a path. Then there exist some $0 \leq i < j \leq d$ such that $x_i = x_j$. But then

$$x_0, x_1, \dots, x_{i-1}, x_i, x_{j+1}, \dots, x_d$$

is an xy trail that is shorter than the original one. This is a contradiction, so this path is a trail. □

Problem 11.16. Prove the Theorem 11.16 using induction.

Problem 11.17. Prove the following. Any odd circuit in a graph has an cycle as a subgraph.

Definition 11.17. The *girth* of a graph is the minimum girth of all cycles in the graph.

Problem 11.18. What is the girth of the cube Q_n? ..of the Peteresen graph?

Problem 11.19. Show that for any vertex u and any other vertex v of the Petersen graph P, there is an *automorphism* f (that is, an isomorphism $f : P \to P$) such that $f(u) = v$. Graphs with this property are called *vertex transitive.*

11.3.1 Distance

Definition 11.18. The *distance*, $\mathrm{dist}(u, v)$ between two vertices u and v in a graph G is the length of the shortest uv-walk in G. The *diameter* $\mathrm{diam}(G)$ of G is the maximum $\mathrm{dist}(u, v)$ over all pairs u and v of vertices in G. The *radius* $\mathrm{rad}(G)$ of G is

$$\min_{u \in V(G)} \max_{v \in V(G)} \mathrm{dist}(u, v).$$

A vertex u that minimises $\max_{v \in V(G)} \mathrm{dist}(u, v)$ is called central.

Problem 11.20. What is the diameter of K_n, C_n, Q_n?

Problem 11.21. Show that for a graph G containing a cycle, the girth of G is at most $2\mathrm{diam}(G) + 1$.

Problem 11.22. How many vertices can a graph of radius k and maximum degree Δ have?

11.3.2 Connected Graphs

Definition 11.19. A graph G is called *connected* if there is an xy-path for every two vertices $x, y \in V(G)$.

Any *maximal* connected subgraph of G, that is, a connected subgraph that is not a proper subgraph of any other subgraph of G, is called a component of G.

Note

The word 'maximal' is different from 'maximum' and is not really used outside of mathematics. The word maximal is defined with respect to a *partial ordering*. (See Section 7.3 of the text for more about partial orders.) An element in a partial ordered set is *maximal* if it is greater than all elements with which it is comparible.
For graphs there are several different partial orders that we might talk about consider.

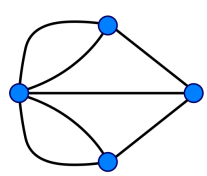
Generally, when we say 'maximal' we are talking about the partial order 'subset inclusion' a graph is maximal in a family of graphs if it is not a subgraph of any graph in the family but itself.

Example 11.20. Let G be the graph containing a K_4 on the vertices $\{1, 2, 3, 4\}$ and a P_2 on the vertices $\{a, b, c\}$. G has two components, K_4 and P_2.

11.3.3 Eulerian Graphs

It is often told that the first problem of graph theory was the problem of the Bridges of Königsberg. In the fictional city of Königsberg (Problem: prove or disprove my unpopular claim that Königsberg is fictional.) there were seven bridges. The dandies of the city, on slow Sunday, would walk about the town, playing a riddlish game. Starting whereever they pleased, they were to walk around, crossing every bridge, without crossing any bridge more then once. This game went on for centuries, and several liars claimed to have done it. Fermat once wrote in the margin of a book that it was simple enough. Until eventually, Euler proved it was impossible, and stabbed Fermat in a dual. (Prove that any of this is true.)

The city of Königsburg consisted of four dots, connected by seven bridges. It looked a lot like this:



How Euler proved it is lost in the depths of historical fiction. But our story inspires the following definitions.

Definition 11.21. An edge-bijective trail in G is called an *euler trail*; an edge-bijective circuit is called an *euler circuit*. A graph is called *eulerian* if it contains an euler circuit.

It is clear that a graph that is disconnected, with edges in each component, cannot be eulerian.

The following is what Euler claims to have proved.

Theorem 11.22. *A connected graph is eulerian if and only if all of its vertices have even degree.*

Proof. That an eulerian graph has all vertices of even degree is clear. Let $C = v_1 v_2 \dots v_N$ be an eulerian circuit. For any vertex v, count the number of times d that v occurs in $v_1 v_2 \dots v_{N-1}$. For each occurence $v = v_i$ is in two edges in C: $v_{i-1}v$ and vv_{i+1}. So v is in $2d$ edges in C. But C is edge-bijectve, so v is in $2d$ edges in G.

Now, in the other direction, assume that G is a connected graph in which every vertex has even degree. We must show that there is an euler circuit in G. There is an edge-bijective trail, indeed any length 1 trail is edge injective, so let $T = v_1 v_2 \dots v_m$ be a longest edge-injective trail in G. First observe that T must be closed. Indeed, if it weren't, then the degree of v_m in G is odd by our proof above, so there is another edge in G incident to $T(v_m)$. Adding this to T, we would get a longer edge-injective trail. So T is an edge-injective circuit.

If T is not edge-bijective, then $G' = G \setminus T$ is non-empty. As G is connected, some edge $e = u_1 u_2 \in G'$ has at least one endpoint $u_1 = v_i$ in $V(T)$. Letting $T' = u_1 u_2 \dots u_{m'}$ be a maximal trail in G' that begins with $u_1 u_2$ we again get that T' must be closed; that is $u_{m'} = u_1 = v_i$. But then

$$v_1 v_2 \dots, v_i u_1 u_2 \dots u_{m'} v_{i+1} \dots v_m$$

is an injective trail in G that is longer than T. This is a contradiction, so T must have been edge-bijective. Thus G is eulerian. □

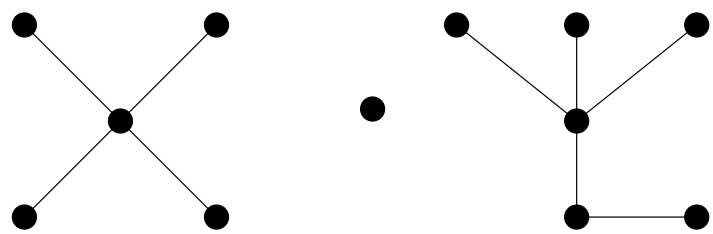
Problem 11.23. Give necessary and sufficient conditions for a graph to have an euler trail. Does Königsberg satisfy these conditions?

11.4 Trees

This is from Section 12.1 of the text. I like it better here.

Definition 11.23. A *forest* is a graph with no cycles as subgraphs. A *tree* is a connected forest.

The following graph is a forest, each of its three components is a tree.



Definition 11.24. A *spanning tree* of a graph G is a spanning subgraph which is a tree.

Theorem 11.25. *A graph G is connected if and only if it has a spanning tree.*

Proof. Let G be a graph with a spanning tree T. Let x and y be any two vertices in G. As T is spanning, they are also in T. As T is connected, there is an xy-path $P = e_1e_2 \dots e_d$ in T. But $E(T) \subset E(G)$ so P is in G as well, that is, it is an xy-path in G. This ways for any $x, y \in V(G)$, so G is connected.

On the other hand assume that G is connected. Let H be a minimal connected spanning subgraph of G, that is, a connected spanning subgraph, which itself has no connected spanning proper subgraph. We show by contradiction that H is a tree. Assume that it isn't. Then it contains a cycle $v_1v_2 \dots v_d$. Let $H' = H \setminus v_1v_2$. If H' is disconnected, then there exist vertices x and y of H such that there are no xy-paths in H'. So all xy-paths in H use v_1v_2, that is, they are of the form $x = x_0x_1 \dots x_iv_1v_2x_{i+1} \dots x_m = y$. But then $x_0x_1 \dots x_iv_1v_nv_{n-1} \dots v_2x_{i+1} \dots x_m$ is an xy-walk in H', contradicting the fact that there is no xy-path in H'. Thus H' is a smaller connected spanning subgraph, contradicting the choice of H. So by contradiction H is a tree. □

Definition 11.26. A *leaf* or *pendant edge* of a tree is a vertex of degree one.

Lemma 11.27. *Any tree has a leaf.*

Proof. Let G be a graph with no leaves. We show that G has a cycle, so is not a tree. This gives the contrapositive of the Theorem. Let $P = v_1v_2 \dots v_n$ be a longest path in G. As v_1 has some neighbour other than v_2 and there is no longer path than P, v_1 has a neighbour v_d on P, so $v_1v_2, \dots v_dv_1$ is a cycle in G. □

Theorem 11.28. *For a tree T, $|V(T)| = |E(T)| + 1$.*

Proof. The proof is by induction on $|V(T)|$. Since the only tree on one vertex is K_1 and the theorem is true for K_1 we assume that $n \geq 2$, and that the theorem is true for graphs on less than n vertices.

As T is a tree, it has a leaf v. Let $T' = T \setminus v$, the graph we get from T by removing v.

Claim 11.29. *T' is a tree*

Proof of claim. To see that it is connected, observe that the only paths in T that contain v, contain v as an endpoint. So for any vertices $x, y \in V(T) \setminus \{v\}$, the xy-path connecting them in T is also in T'. As $T' \leq T$ it has no cycles. So T' is a tree. $\diamond$

Now $|V(T)| - 1 = |V(T')|$, which by induction is $|E(T')| + 1 = |E(T)|$. So $|V(T)| = |E(T)| + 1$ as needed. $\square$

Definition 11.30. A *cut edge* of a graph G is any edge whose removal increases the number of components of G.

Theorem 11.31. *Let $G = (V, E)$ be a graph. Then the following are equivalent.*

i. G is a tree.

ii. G is connected, but every edge is a cut edge.

iii. G has no cycles, and $|V| = |E| + 1$.

iv. G is connected, and $|V| = |E| + 1$.

v. Any two vertices of G are connected by a unique path.

Proof. From the previous theorem we have that (i) implies (iii) and (iv). We show that (iii) $\Rightarrow$ (i) and (iv) $\Rightarrow$ (ii) $\Rightarrow$ (i). This shows the equivalence of the first four statements. As an exercise, show that statement (v) is also equivalent.

(iii) $\Rightarrow$ (i): Assume that G has no cycles, so each component of G is a tree. For each component C we thus have that $|V(C)| = |E(C)| + 1$. So $|V(G)| = |V(E)| + c$ where c is the number of components. By $|V| = |E| + 1$, there is only one component, so it is a tree.

(iv) $\Rightarrow$ (ii): Assume that G is connected an has $|V| = |E| + 1$. If there is some edge e that is not a cut-edge, then $G' = G \setminus \{e\}$, the graph we get by removing e, is connected, and so contains a spanning tree T. Thus $|V| = |V(G')| = |V(T)| = |E(T)| + 1 \leq |E(G')| + 1 = |E| + 2$, but this is a contradiction.

(ii) $\Rightarrow$ (i): Let G be connected and every edge of G be a cut-edge. As G is connected it has a spanning tree. Since every edge is a cut edge, no proper subgraph has a spanning tree, so G itself is a spanning tree.

□

11.5 Bipartite and Planar Graphs

11.5.1 Bipartite graphs

Definition 11.32. A graph G is *bipartite* if and only if there is a partition $A \cup B = V(G)$ of its vertices (so $A \cap B = \emptyset$), such that all edges are between A and B, (ie, that A and B both induce empty sets.

Problem 11.24. Show that if G is bipartite if and only if $G \to K_2$. Show that $C_{2n+1} \to G$ implies that G is not bipartite.

We can say more

Proposition 11.33. *For a graph G, the following are equivalent*

i. *G is bipartite.*

ii. *$G \to K_2$.*

iii. *G contains no odd cycles.*

iv. *$C_{2n+1} \not\to G$ for all integers $n \geq 1$*

Proof. We've proved most of this over the course of these notes. With impliction iii) $\Rightarrow$ ii), the proof can be pieced together from the exercises. So we prove this implication.

Assume that G has no odd cycles. We define a homomorphism $f : G \to K_2$ as follows. Let v_0 be some vertex in G. For every vertex v in G let

$$f(v) = \begin{cases} 0 \text{ if } \mathrm{dist}(v, v_0) \text{ is even} \\ 1 \text{ if } \mathrm{dist}(v, v_0) \text{ is odd} \end{cases}$$

We claim that this is a homomorphism. If it is not, there are some two vertices adjacent u and v such that $f(u) = f(v)$. That is, $\mathrm{dist}(u, v_0)$ and $\mathrm{dist}(v, v_0)$ have the same parity. But then there is path P_1 from v_0 to v and P_2 from u to v_0 of the same parity, so the path $P_1 + vu + P_2$ is an odd path from v_0 to v_0. That is, it is an odd circuit in G. But we showed this implies there is an odd cycle in G, which contradicts our assumption. □

Problem 11.25. Let G be a graph with n vertices and $m > (n/2)^2$ edges. Show that G cannot be bipartite.

11.5.2 Planar Graphs

Can we draw the petersen graph in the plane so that no edges cross?

A graph G is *planar* if it can be drawn in the plane with no edges crossing. A drawing of the graph in the plane with no edges crossing is called an *(planar) embedding* of G.

Problem 11.26. Show that K_4 is planar but that K_5 is not.

We can show that K_5 is not planar by ad-hoc arguments, but such arguments difficult for larger graphs. That's okay though. That K_5 is not planar tells us that a lot of other graphs are not planar.

Problem 11.27. Show that no graph containing K_5 as a subgraph, can be planar. Show that the petersen graph is not planar.

Definition 11.34. A *subdivision* of a graph G is a graph we get from G by dividing edges. That is, G' is a subdivision of G if we can construct it by repeated application of the following alteration, called an *edge subdivision*.

- remove an edge $e = ab$,
- add a new vertex v_e,
- add the edges av_e and bv_e.

It should be pretty clear that because K_5 is not planar, no subdivision of K_5 is planar. So no graph containing a subdivision of K_5 as a subgraph is planar. Can you think of any other graphs that are not planar? Please don't look ahead. It will ruin the surprise.

Okay. Here's the surprise. $K_{3,3}$ is also not planar, and has no subdivision of K_5. But that is essentially it. Kuratowski showed the following.

Theorem 11.35 (Kuratowski's Theorem)**.** *A graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$ as a subgraph.*

We won't prove this, as it is a little involved. But we should show the following.

Problem 11.28. Show that $K_{3,3}$ is not planar.

One can do this with ad-hoc arguments. But we can also do it more methodically.

11.5.3 Euler's Formula

A planar embedding of a graph G defines *faces* in the plane: the little bits of plane you get by cutting along all the edges of the embedded graph. Where $|V(G)| = n$ and $|E(G)| = m$ and f is the number of faces. Euler showed the following fundamental identity.

Theorem 11.36 (Euler's Formula)**.** *For any embedding of a connected graph G in the plane,*

$$n - m + f = 2.$$

Proof. The proof is by induction on m. As G is connected, $m \geq n - 1$. If $m = n - 1$ then G is a tree, so there is only one face, and the identity holds. This is our base case.

Now assume that $m \geq n$. Then there is an non-cut edge e. Remove it. Its removal joins two faces, so reduces f by one, but also reduces e by one. So the identity holds from the induction hypothesis. □

Corollary 11.37. *If G is connected and planar and $n \geq 3$, then*

$$m \leq 3n - 6.$$

Proof. Count the number N of pairs (e, r) where e is an edge, and r is a face with e on the boundary. As every edge is in at most 2 faces, we have that $N < 2m$ and as every face has at least 3 edges, $3f < N$ so

$$f \leq N/3 \leq \frac{2}{3}m.$$

Plugging this into $n - m + f = 2$ gives that

$$n - m/3 = n - m + \frac{2}{3}m \geq 2$$

so $m \leq 3n - 6$ as needed. □

Problem 11.29. Use this corollary to show that K_5 and $K_{3,3}$ are not planar.

Problem 11.30. ?? Show that any planar graph has a vertex of degree at most 5.

11.5.4 Dual Graphs

Definition 11.38. Given an embedding of a graph G we construct a new (multi-)graph G^*, called the *dual of G,* from G as follows.

The vertices of G^* are

$$V(G^*) = \{v_f \mid f \text{ is a face of } G\ \}.$$

The edges are defined as follows. For every edge e of G let $e^* = v_f v_g$, where f and g are the faces of G incident to e. Then $E(G^*)$ is the **multiset**

$$E(G^*) = \{e^* \mid e \in E(G)\}.$$

(Note that both $V(G^*)$ and $E(G^*)$ depend on the particular embedding we chose for G.)

Not only is G^* a multigraph, it can have loops:

Problem 11.31. What is the dual of a tree on n vertices.

Problem 11.32. Find conditions on a graph G so that its dual G^* has no loops. Find conditions on G so that its dual, under any embedding, has no multiedges.

Problem 11.33. Give an embedding of K_4 and construct its dual. Can you find an embedding of K_4 that gives a different dual? Can you find some graph G with two different embeddings that yield different duals?

Problem 11.34. Show that the dual of an embedded graph is planar.

Theorem 11.39. *Let G be planar and G^* the dual of an embedding of G. Then G is bipartite if and only if G^* is eulerian.*

Proof. One way is easy. Let G be a bipartite graph embedded in the plane. For any vertex v_f of G^*, the walk around the boundary of f is a closed walk. As G is bipartite, it is even, so v_f has even degree. This was for any vertex of G^*, so G^* is euleriain.

On the other hand, assume that G^* is eulerian. We want to show that every cycle of G is even. (This is clear of the boundary of faces under a given embedding but not of every cycle.) Fix an embedding of G.

Let C be a cycle in G, and let $A \subset F(G)$ be the set of faces inside C relative to the embedding. Then C is the symmetric difference of the edge sets of faces in A. Each face has a even number of edges, so so does their symmetric difference. Thus C is even. □

Problem 11.35. Draw an embedding of Q_3. Draw its dual.

11.6 Hamilton Paths and Cycles

Definition 11.40. A path, or cycle, in G is *hamilton* if it is bijective. A graph G is *hamiltonian* if it contains a hamiltonian cycle.

There were some nice conditions that characterised the eulerian graphs. We are not so lucky with hamilton graphs. In fact, the problem of deciding if a graph on n vertices is hamiltonian, is NP-complete. This means the best known algorithm we have for deciding, takes time that is exponential in in n, (at least for some graphs). This doesn't mean we can't say anything about graphs that are hamiltonian. Indeed they must be connected, and have no leaves. We can say a bit more.

Problem 11.36. Find a hamilton path in Q_n.

Problem 11.37. How many hamilton cycles are there in the n-wheel W_n. The W_n is the n-cycle C_n with one more vertex h adjacent to all other vertices.

Here is an early result of Dirac.

Theorem 11.41 (Dirac 1952)**.** *A graph G with n vertices is hamiltonian if $\delta(G) \geq n/2$.*

Instead of proving this we prove a slightly stronger result of Ore.

Theorem 11.42 (Ore 1960)**.** *A graph G with $n \geq 3$ vertices if hamiltonian if for every two vertices $x \neq y$ with $x \not\sim y$, we have $\deg(x) + \deg(y) \geq n$.*

Proof. Towards contradiction, let G be a maximal counterexample on $n \geq 3$ vertices. So G is not hamiltonian, and satisfies

$$(*) \qquad x \not\sim y \Rightarrow \deg(x) + \deg(y) \geq n,$$

but $G + xy$ is hamiltonian for any $x \not\sim y$. (We may assume that G is not complete, as complete graphs are clearly hamiltonian.) Let $H = G + xy$ for some $x \not\sim y$ in G. Then H is hamiltonian, and any Hamilton cycle contains xy. So there is a hamilton path

$$x = x_1 \sim x_2 \sim \ldots x_n = y.$$

Now if $x_1 \sim x_i$ and $x_{i-1} \sim x_n$, then

$$x_1 \sim x_2 \sim \cdots \sim x_{i-1} \sim x_n \sim x_{n-1} \sim \cdots \sim x_i \sim x_1$$

is a hamilton cycle not using xy so in G, which is a contraction. So for each of x_1s $\deg(x_1)$ neighbours x_i in $\{x_2, \ldots, x_{n-1}\}$ (recall that $x_1 \not\sim x_n$), x_{i-1} is not a neighbour of x_n. So

$$\deg(x) + \deg(y) = \deg(x_1) + \deg(x_n) \leq \deg(x_1) + (n - 1 - \deg(x_1)) \leq n - 1$$

This contradicts $(*)$, so the theorem is true.

□

Problem 11.38. Prove Diracs theorem using Ore's theorem.

Problem 11.39. Do problems 10, 18, 21 from Section 11.5 of the text.

11.7 Graph Colouring

Definition 11.43. A *(proper) k-colouring* of a graph G is a function $f : V(G) \to [k]$ such that

$$u \sim v \Rightarrow f(u) \neq f(u).$$

If G has a k-colouring, it is *k-colourable*. The *chromatic number* $\chi(G)$ of G is the minimum k for which G is k-colourable. If $k = \chi(G)$, we also say that G is *k-chromatic*.

Problem 11.40. For what graph H is the following statement true? A graph G is k colourable if and only if $G \to H$.

Problem 11.41. What is the chromatic number of:

i. K_n

ii. C_n

iii. Q_n

iv. W_n

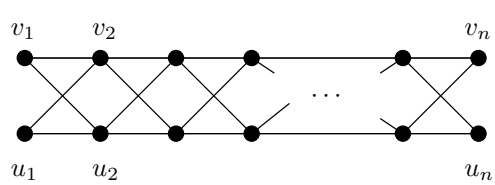
v. The petersen graph.

Problem 11.42. Describe all 2-colourable graphs. (That is, complete the statement " A graph is 2-colourable if and only if ... " using properties we've seen before.)

Problem 11.43. (This is hard.) Let $G(n)$ be the graph (pictured below) with vertices $V(G) = \{v_1, \ldots, v_n\} \cup \{u_1, \ldots, u_n\}$ and the following edges.

- $v_i \sim v_j$ if $|i - j| = 1$.
- $u_i \sim v_j$ if $|i - j| = 1$.
- $u_i \sim u_j$ if $|i - i| = 1$.

How many 3-colourings does the following graph G have? (A recursive answer is fine.)



Definition 11.44. The *clique number* $\alpha(G)$ of a graph G is maximum n such that K_n is a subgraph of G.

Problem 11.44. Show that $\chi(G) \geq |V(G)|/\alpha(\overline{G})$.

Definition 11.45. A graph G is k-critical if it is k-chromatic, but any proper subgraph is $(k-1)$-colourable.

Problem 11.45. Show that K_k is k-critical.

Problem 11.46. Show that any k-chromatic graph contains a k-critical graph as a subgraph.

Problem 11.47. Find all 3-critical graphs.

Problem 11.48. Show that any n-critical graph G has $\delta(G) \geq n - 1$.

From the last two problems and Problem **??** you should be able to do the following.

Problem 11.49. Show that any planar graph is 6-colourable.

In fact, more is true:

Theorem 11.46. *(The Four-Colour Theorem [Haken-Appel 1976]) Any planar graph is* 4*-colourable.*

The proof of this is very long and difficuly. But it is an interesting exercise (maybe also a bit difficult) to prove that every planar graph is 5-colourable.